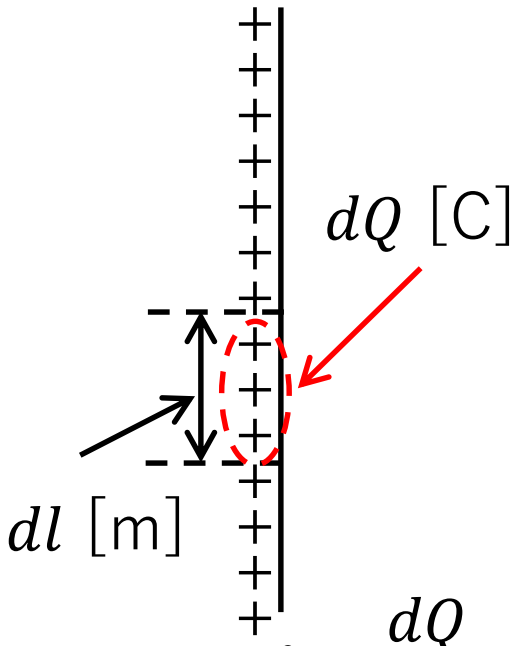
分布電荷による電界

電荷の分布

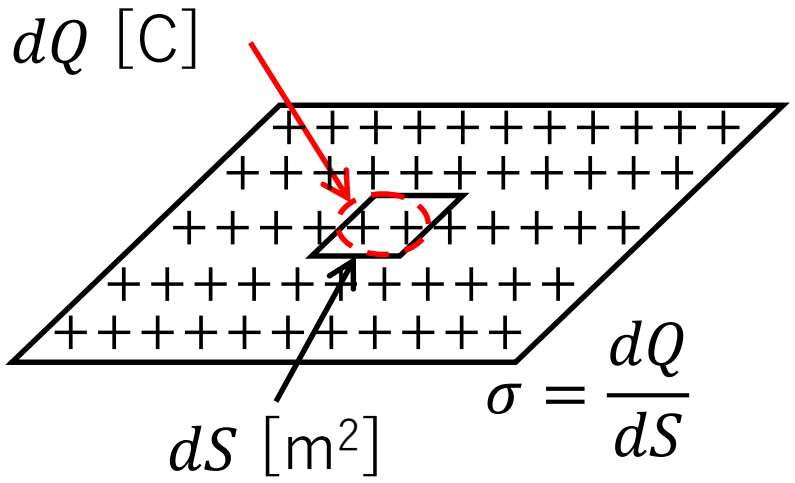
線電荷密度：λ [C/m]
(長さ1 [m]あたりにλ [C])

面電荷密度：σ [C/m^2]
(面積1 [m^2]あたりにσ [C])

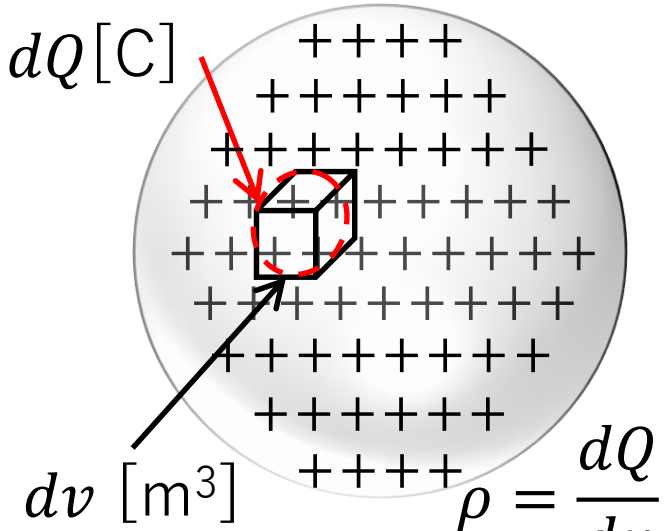
体積電荷密度：ρ[C/m^3]
(体積1 [m^3]あたりにρ [C])



$$\lambda = \frac{dQ}{dl}$$



$$\sigma = \frac{dQ}{dS}$$



$$\rho = \frac{dQ}{dv}$$

ガウスの法則に適応する場合

$$Q = \int_l \lambda dl, \qquad Q = \int_S \sigma dS, \qquad Q = \int_V \rho dv$$